

Yash Kaul

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

Master of Science in Computer Science - GPA: 4.00/4.00

Jan 2024 – May 2025

- Relevant Coursework: Database Systems, Distributed Systems and Algorithms, Graph Mining, Parallel Computing, Network Programming

Rensselaer Polytechnic Institute

Troy, NY

Bachelor of Science in Computer Science - GPA: 3.73/4.00

Aug 2021 – May 2024

- Relevant Coursework: Data Structures and Algorithms, Advanced Algorithms, Linear Algebra, Graph Theory, Operating Systems, Numerical Analysis, Programming Languages, Software Design and Documentation
- Course Assistant for Data Structures and Algorithms:
Host lab/office hours, review sessions, and grade exams for 300+ students over six semesters
- Member of Upsilon Pi Epsilon (International Honors Society for Computing and Information Disciplines)

EXPERIENCE

Software Engineering Intern

Remote

IoT4AI

May 2024 - Aug 2024

- Developed simulation tool testing latency and availability of IoT devices using React, Three.js, and NumPy
- Created dynamic shortest path algorithm for real-time networks, decreasing operational latency by 43%
- Built data-streaming pipeline processing updates from 1000+ BLE microcontrollers using RabbitMQ and Redis
- Designed caching system to store the state of the network over time—empowering fluid visual network changes

Undergraduate Research Assistant

Troy, NY

Rensselaer Polytechnic Institute

Aug 2023 - May 2024

- Implemented bi-connectivity algorithm with distributed messaging and shared-memory in OpenMPI and C++

Software Engineering Intern

Remote

CollabLab

May 2023 - Dec 2023

- Developed robust cookie-based JWT authentication/login systems, improving security of sensitive user data
- Designed middleware architecture using Passport.js; incorporating OAuth 2.0 (Google, Facebook, GitHub, etc.)
- Spearheaded infrastructure solution using CloudFormation: setup DNS, VPC subnets, Application Load Balancer, and ECS Clusters—cutting server costs by 31% and streamlining deployment workflow
- Automated deployment of microservices using Docker and ECR—enabling Zero Downtime Deployment

PROJECTS

Art2Ascii | TypeScript, VSCodeAPI, Sharp, Esbuild | [link](#)

May 2024 – Present

- Created VSCode extension to play ASCIIified GIF's in Explorer panel using Sharp for image processing
- Implemented color search algorithm using k-d trees, resulting in 10x speedup on image generation
- Published on Microsoft's Visual Studio Marketplace with 80+ users

PictoPlan | Python, FastAPI, React, MongoDB, OpenAI API, Google Tesseract | [link](#)

Nov 2023 – Jan 2024

HackRPI Best First Time Hack Winner

- Launched web-platform for educators to create guides from visual content using Google Tesseract and OpenAI API
- Designed data pipeline to store in MongoDB and then process images into text and summarize using GPT-3.5
- Collaborated with four colleagues over the course of 24 hours and deployed using Heroku and Cloudflare

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript/TypeScript, HTML/CSS, MATLAB

Software: PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, RabbitMQ, Kafka, Express, React, Node.js, Tailwind, Three.js, NumPy, FastAPI, Git, OpenMPI, JUnit, GDB, Valgrind, Postman

Cloud: API Gateway, S3, ECR, DNS, AWS SQS, AWS ECS/EKS, AWS Lambda, CloudFormation, CodeDeploy

Certifications: [AWS Cloud Practitioner](#)